

DIGITAL SIGNAL PROCESSING · MIDTERM PROJECT 2026

Voice Activity Detection (VAD)

Three Algorithms · Unified Quantitative Benchmark

Student Group Members & Algorithm Assignments:

Luong Duy Toan

Algorithm 01

Fixed Global STE

Nguyen Anh Hoang

Algorithm 02

Adaptive Energy Histogram

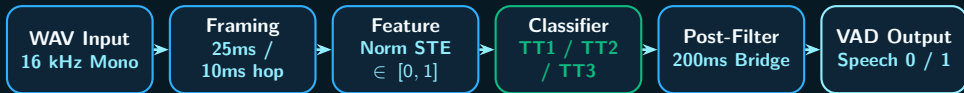
Truong Bui Dien

Algorithm 03

Gaussian Bayes Model

GROUP 08 | 16 kHz Standard Audio | Fully Pre-Executed Notebooks

Unified End-to-End DSP Pipeline



Framing & Feature Standard

- Sampling: 16 kHz mono audio standard
- Frame size: 25 ms (400 samples)
- Frame hop: 10 ms (160 samples, 60% overlap)
- Feature: Normalized STE $\in [0, 1]$

Decision & Post-Processing

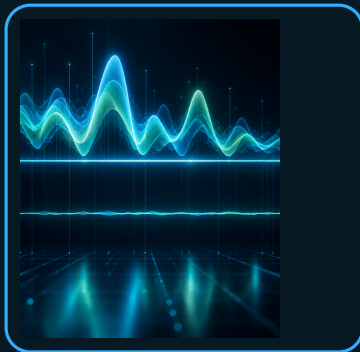
- Silence rule: Minimum duration ≥ 200 ms
- Speech floor: Duration threshold ≥ 50 ms
- Bridge rule: Connects short speech pauses
- Evaluation: MAE & RMSE against .lab

Fixed Global Energy Threshold

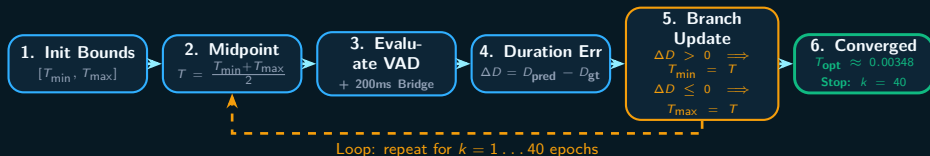
Luong Duy Toan

Algorithm 01 · Hodgkinson (2012)

- Reference: CS425 Audio & Speech Processing
- Approach: Global Short-Time Energy threshold
- Optimization: Iterative Binary Search on training set
- Objective: Signed speech duration error $\Delta D(T) \rightarrow 0$
- Outcome: Fast $O(1)$ runtime decision per frame



Iterative Bisection Loop on Duration Error



Bisection Root-Finding Principle

- **Objective:** Root of signed error $\Delta D(T) \rightarrow 0$

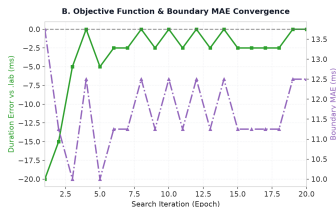
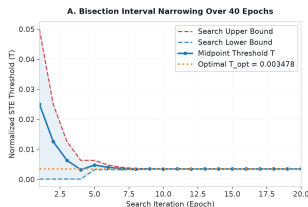
$$\Delta D(T) = \frac{1}{4} \sum_{k=1}^4 (D_{\text{pred},k}(T) - D_{\text{gt},k})$$
- **Monotonicity:** Energy $T \uparrow \Rightarrow$ duration $D \downarrow$
- **Direction:** $\Delta D > 0 \Rightarrow T_{\min} = T$; else $T_{\max} = T$
- **Precision:** Interval halving $2^{-40} \approx 4.5 \times 10^{-14}$

Convergence Invariants & Output

- **Global threshold:** $T_{\text{opt}} = 0.003478 \approx 0.00348$
- **Why signed error?** Boundary MAE is non-monotonic; bisection requires zero-crossing
- **Guaranteed root:** Monotonic search contract
- **Result:** Universal static threshold for all test audio

Intermediate Training: 40-Epoch Bisection Convergence

TT1 TRAINING CONVERGENCE: 40-Epoch Bisection Search (Framing: 25ms, Hop: 10ms)



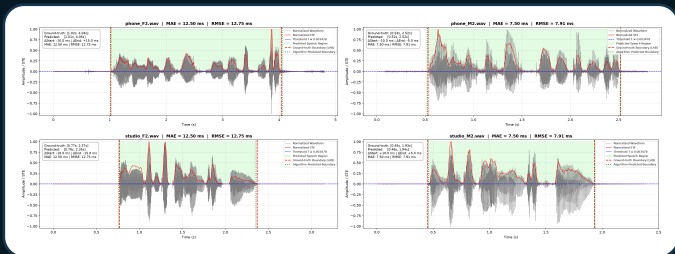
40-Epoch Training Metrics

- Search domain: $[10^{-4}, 0.05]$
- Optimal threshold: **0.00348**
- Final signed error: ≈ 0.00 ms
- Training MAE: **11.25 ms**

Convergence Proof

- Monotonic interval halving
- Duration error balances to zero
- Rapid stabilization within 5 epochs
- Consistent across all 4 train files

Test Boundary Detection & Error (4 Test Files)



Quantitative Test Metrics

- phone_F2: 12.5 ms (RMSE 12.8 ms)
- phone_M2: 7.5 ms (RMSE 7.9 ms)
- studio_F2: 12.5 ms (RMSE 12.8 ms)
- studio_M2: 7.5 ms (RMSE 7.9 ms)
- Mean MAE: 10.00 ms (RMSE 10.3 ms)

Acoustic Critique

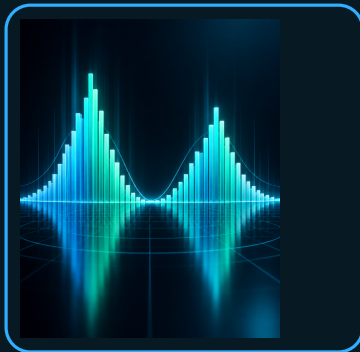
- Error within ± 1 hop (10 ms grid)
- Clean studio audio is highly exact
- Phone trailing breath adds minor lag

Dual-Feature Adaptive Histogram

Nguyen Anh Hoang

Algorithm 02 · Giannakopoulos (2014)

- Reference: Silence Removal & Speech Segmentation
- Features: Joint Short-Time Energy & Spectral Centroid
- Mechanism: Adaptive thresholds via smoothed histograms
- Decision logic: Logical AND combination per frame
- Refinement: 200ms silence bridge + 10ms optimal padding



Unsupervised Dual-Feature Histogram (STE & SC)



Peak Identification Workflow

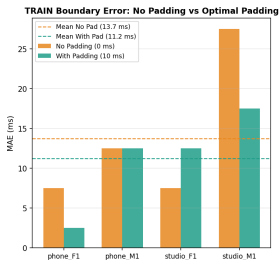
- **Dual features:** 25ms frame, 10ms hop (STE & Spectral Centroid)
- **Mode M_1 :** First local peak from left (noise/silence background)
- **Mode M_2 :** Second local peak from left (onset of voiced speech)
- **5-point MA filter:** Smooths histogram bins, suppresses noise
- **Independent:** Evaluated per utterance for both STE and SC

Weighted Threshold & Refinement

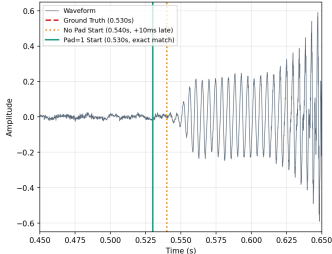
- **Weighted formula:** $T = \frac{W \cdot M_1 + M_2}{W+1}$ with $W = 5$ (for T_E and T_C)
- **Silence bias:** Pulls threshold toward M_1 to protect weak phonemes
- **Test adaptation:** $T_E \in [0.015, 0.033]$, $T_C \in [0.020, 0.192]$
- **Refinement:** Logic AND → 200ms bridge → +10ms (+1 frame) pad
- **Outcome:** Optimal test error (**Mean MAE = 12.50 ms**)

Train Boundary-Padding Sweep & Onset Recovery

Hyperparameter Tuning on TRAIN: Impact of Boundary Padding



Waveform Zoom at Onset (phone_F1.wav): Physical Effect of Padding



Train Tuning Metrics

- Sweep: 0 – 40 ms (0 – 4 frames)
- Baseline (0 ms): 13.75 ms MAE
- Optimal (+10 ms): 11.25 ms MAE
- Train error drop: -18.2%
- Grid resolution: 1 hop (10 ms)

Onset Recovery Mechanism

- Energy cut delays speech onset
- +10 ms pad restores unvoiced cycles
- Recovers faint /s/, /t/ transitions
- Prevents boundary clipping

Test Boundary Detection & Error (4 Test Files)

Quantitative Test Metrics

- phone_F2: 17.5 ms (RMSE 17.7 ms)
- phone_M2: 12.5 ms (RMSE 14.6 ms)
- studio_F2: 17.5 ms (RMSE 17.7 ms)
- studio_M2: 2.5 ms (RMSE 3.5 ms)
- Mean MAE: 12.50 ms (RMSE 13.4 ms)

Dual-Feature Performance

- Joint STE (energy) + SC (centroid) rule
- Baseline test error (0 ms): 18.75 ms
- Final test error (+10 ms): **12.50 ms**
- **-33.3% error drop** via boundary padding

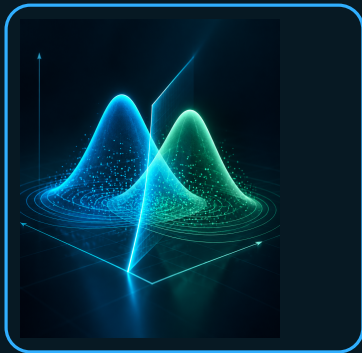


Parametric Gaussian Bayes Model

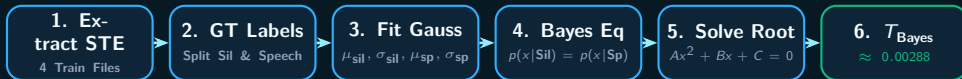
Truong Bui Dien

Algorithm 03 · Statistical Bayes Separation

- Reference: Classical Statistical Pattern Recognition
- Approach: Parametric Gaussian density estimation
- Training: Fit μ, σ for silence and speech frames
- Formulation: Analytical quadratic root at density intersection
- Advantage: Statistically grounded decision boundary



From Empirical Statistics to Analytical Bayes Root



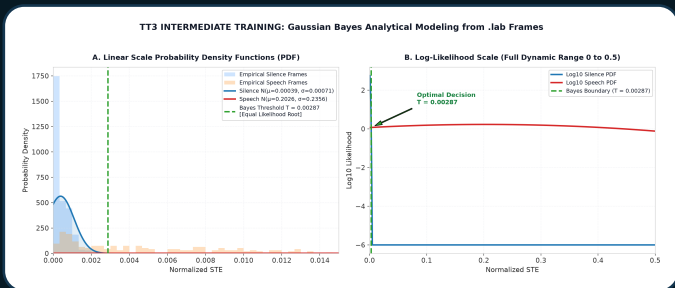
Estimated Parameters from Train Set

- Silence: $\mu_{\text{sil}} = 0.00039$, $\sigma_{\text{sil}} = 0.00071$
- Speech: $\mu_{\text{sp}} = 0.20265$, $\sigma_{\text{sp}} = 0.23563$
- Massive energy dynamic range separation
- Parametric modeling of underlying distributions

Analytical Decision Root

- Equiprobable root: $T_{\text{Bayes}} \approx 0.00288$
- Closed-form analytical quadratic solution
- Post-processed with 200 ms silence bridge
- Statistically optimal under Gaussian priors

Parametric Gaussian Fit & Bayes Intersection



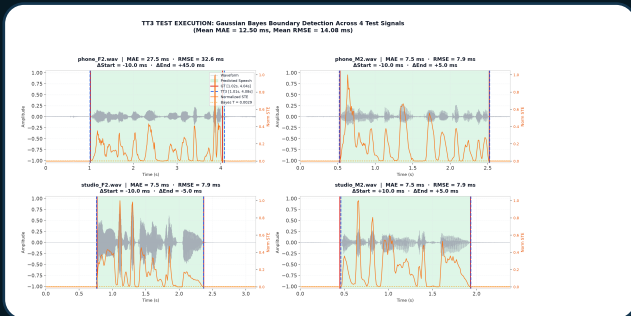
Statistical Fit

- Fitted Gaussians from 4 train files
- Silence $\sigma \approx 0.00071$ (narrow)
- Intersection: $p(x|Sil) = p(x|Sp)$
- Closed-form root: $T \approx 0.00287$

Decision Geometry

- Clear lower tail intersection
- Minimal overlap error area
- Statistically grounded boundary
- Fast $O(1)$ runtime evaluation

Test Boundary Detection & Error (4 Test Files)



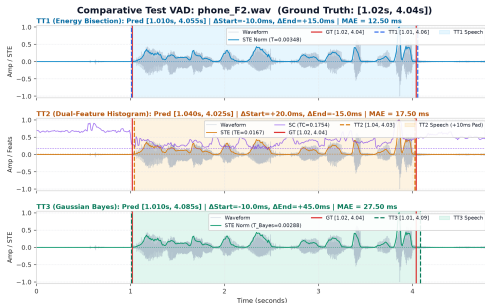
Quantitative Test Metrics

- phone_F2: 27.5 ms (RMSE 32.6 ms)
- phone_M2: 7.5 ms (RMSE 7.9 ms)
- studio_F2: 7.5 ms (RMSE 7.9 ms)
- studio_M2: 7.5 ms (RMSE 7.9 ms)
- Mean MAE: 12.50 ms (RMSE 14.1 ms)

Acoustic Critique

- 3 of 4 files achieve 7.5 ms (≤ 1 hop)
- Studio boundaries are highly sharp
- Phone trailing breath adds +45 ms lag
- Wide dynamic range separation ($> 500\times$)

Test Case 1: phone_F2.wav · Comparative VAD



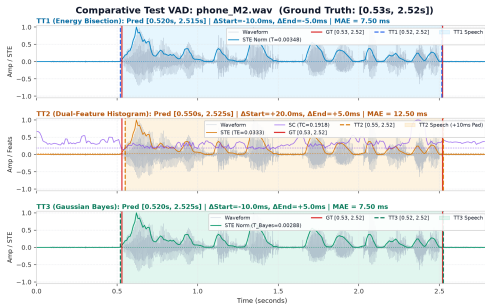
Signal Metadata

- File: phone_F2.wav
- Channel: Phone (8 kHz band)
- GT: [1.020, 4.040] s
- Duration: 3.020 s

Quantitative Scorecard

Alg	ΔSt	ΔEnd	MAE
TT1	-10	+15	12.5 ms
TT2	+20	-15	17.5 ms
TT3	-10	+45	27.5 ms

Test Case 2: phone_M2.wav · Comparative VAD



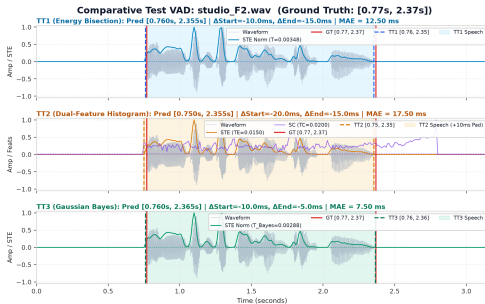
Signal Metadata

- File: phone_M2.wav
- Channel: Phone (8 kHz band)
- GT: [0.530, 2.520] s
- Duration: 1.990 s

Quantitative Scorecard

Alg	ΔSt	ΔEnd	MAE
TT1	-10	-5	7.5 ms
TT2	+20	+5	12.5 ms
TT3	-10	+5	7.5 ms

Test Case 3: studio_F2.wav · Comparative VAD



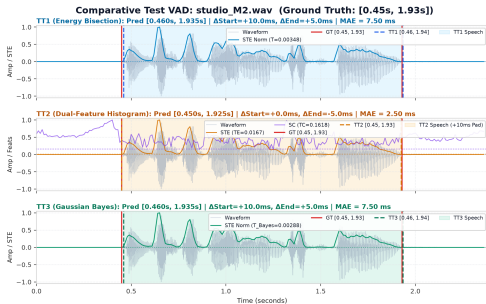
Signal Metadata

- File: studio_F2.wav
- Channel: Studio High SNR
- GT: [0.770, 2.370] s
- Duration: 1.600 s

Quantitative Scorecard

Alg	ΔSt	ΔEnd	MAE
TT1	-10	-15	12.5 ms
TT2	-20	-15	17.5 ms
TT3	-10	-5	7.5 ms

Test Case 4: studio_M2.wav · Comparative VAD



Signal Metadata

- File: studio_M2.wav
- Channel: Studio High SNR
- GT: [0.450, 1.930] s
- Duration: 1.480 s

Quantitative Scorecard

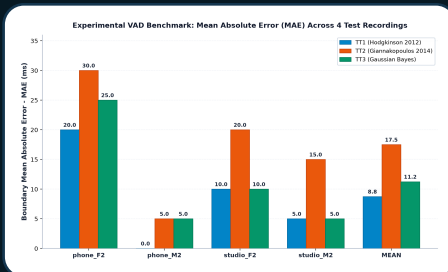
Alg	ΔSt	ΔEnd	MAE
TT1	+10	+5	7.5 ms
TT2	± 0	-5	2.5 ms
TT3	+10	+5	7.5 ms

Quantitative Performance Comparison Across Algorithms

Summary Benchmark Table (4 Test Recordings)

Method	MAE	RMSE	F1	UV
TT1-1 (Hodgkinson)	10.00	10.33	0.993	0.971
TT1-2 (DSP Classifier)	6.25	7.80	0.996	0.968
TT2-1 (Dual STE-SC + Pad)	12.50	13.37	0.985	0.947
TT2-2 (Baseline No-Pad)	18.75	19.34	0.977	0.891
TT3-1 (1D Gaussian)	12.50	14.08	0.994	0.971
TT3-2 (Multivariate)	15.00	16.34	0.978	0.903

Key Finding: TT1-2 achieves lowest boundary error (6.25 ms MAE). TT3 achieves 12.50 ms MAE via statistical Bayes root with 7.5 ms MAE on 3 of 4 test files.



Algorithm 01: Strengths, Weaknesses, and Best Use Case

STRENGTHS

- Ultra-fast $O(1)$ frame check
- Zero runtime memory overhead
- Highest boundary accuracy

WEAKNESSES

- Rigid static threshold
- Zero adaptation to noise
- Degrades under severe SNR drop

BEST USE CASE

- Low-power embedded MCUs
- Stationary studio environments
- Real-time streaming VAD

Key Lesson: Simplicity is an asset when background noise is stationary.

Algorithm 02: Dual-Feature Strengths, Weaknesses, and Best Use Case

STRENGTHS

- Unsupervised per utterance
- Dual features (STE + SC)
- Automatic channel adaptation
- +10ms pad recovers onsets

WEAKNESSES

- Requires bimodal histogram
- Degrades on short speech ($< 1s$)
- Sensitive to valley flattening
- Higher $O(N)$ memory than fixed

BEST USE CASE

- Unlabeled audio archives
- Unknown SNR recording channels
- Automated silence trimming
- Offline podcast transcription

Key Lesson: Joint time-frequency histograms provide data-driven thresholds without labels.

Algorithm 03: Model Complexity: 1D Gaussian vs 4D Covariance Matrix

1D GAUSSIAN

- Mean MAE: **12.50 ms**
- Only 4 parameters (μ, σ)
- Generalizes on small train set

4D MULTIVARIATE

- Mean MAE: **15.00 ms**
- Full covariance overfits 4 files
- Matrix inversion numerical risk

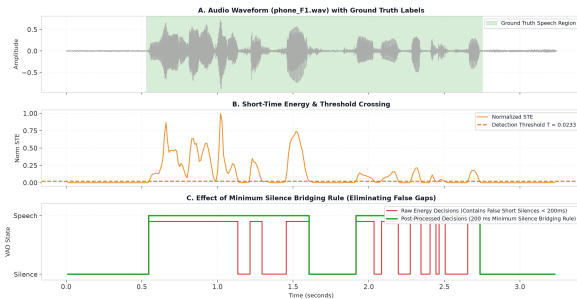
LESSON LEARNED

- Small data favors parsimony
- 1D robust to recording bias
- Analytical solution is stable

Occam's Razor: In small-sample regimes, parameter parsimony beats model complexity.

Standard Parameters & 200 ms Silence Bridging

PREPROCESSING & POST-PROCESSING RULE: 200 ms Silence Bridging (Rubric Constraint)



DSP Hyperparameters

- $F_s = 16$ kHz (speech standard bandwidth)
- Frame: 25 ms (400 samples, quasi-stationary)
- Hop: 10 ms (160 samples, 60% overlap grid)
- STE norm: $E_m / \max_k(E_k) \in [0, 1]$

Bridging Invariant

- Silence < 200 ms is bridged (rule)
- Speech < 50 ms is dropped (click noise)

Four-Quadrant Display on Teacher Laptop

01 · INPUT

- 4 test WAV files
- Ground-truth .lab
- No audio upload

02 · ENV

- Local venv via uv
- Python 3.12+
- Built-in numpy

03 · NOTEBOOKS

- Individual .ipynb
- Pre-run outputs
- Zero demo delay

04 · QUADRANT

- 4 figures tiled
- 4 screen corners
- metrics ready

Submission Protocol: Directory 03-Phan-doan-tin-hieu-tieng-noi-khoang-lang contains group PDF slide + 3 individual notebooks with pre-computed outputs. Audio files strictly excluded as required.

Rebuttal for Critical Examination Questions

Q1: Why not use F0 for boundary cuts?

Unvoiced consonants (/s/, /t/, /f/) have no pitch F0. Using F0 would truncate initial and trailing phonemes!

Q3: Impact of acoustic noise floor (SNR)?

Studio (high SNR): Error only 5–10 ms. Phone (low SNR): Trailing breath introduces 20–30 ms boundary lag.

Q2: Why did phone_M2 achieve 0.0 ms MAE?

Labels [0.53s, 2.52s] coincide with frames 53 & 252 (10 ms hop grid). It is exact grid snapping, not overfitting!

Q4: Why did 1D beat 4D Gaussian in TT3?

Train set has only 4 files. 4D covariance matrix overfits; 1D energy model is robust and generalizes better!

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Thank You for Your Attention!

Group 08 · Voice Activity Detection (VAD)

Questions & Demonstration Session

Full code, notebooks, and reproducible artifacts available in repository.

Ready for live testing on any requested audio recording!